Project Title: Bus Reservation System

1. Introduction

🕐 Purpose: Define the objectives of the Bus Reservation System (BRS), including online booking, seat allocation, cancellation, and schedule management.

🕐 Scope: The system will allow passengers to search buses, reserve seats, make payments, and receive tickets. Admins can manage bus schedules, routes, fares, and reservations.

🕐 Definitions, Acronyms, Abbreviations:

o BRS: Bus Reservation System

o UI: User Interface

o DBMS: Database Management System

🕐 References: IEEE SRS guidelines, institutional project standards, and bus transport policies.

2. Overall Description

🕐 Product Perspective: The system is a web/mobile application integrated with a central database.

🕐 Product Features:

o User registration/login

o Bus search by route/date

o Seat reservation and cancellation

o Online payment gateway

o Ticket generation (PDF/Email/SMS)

o Admin dashboard for bus/fare management

🕐 User Classes and Characteristics:

o Passenger: Books and cancels tickets.

o Admin: Manages buses, routes, and reservations.

o System: Handles seat allocation, payments, and notifications.

🕐 Operating Environment:

o Web application (HTML, CSS, JavaScript, PHP/Java/Python)

o Database (MySQL/PostgreSQL)

o Mobile app (Android/iOS optional)

🕐 Constraints:

o Secure payment handling

o Real-time seat availability

o Compliance with transport regulations

3. System Features

Feature

Description

User Registration

Create/manage user accounts

Priority

High

Bus Search

Search buses by route/date

High

Seat Reservation

Select and book seats

High

Payment Processing Secure online transactions

High

Ticket Generation

Issue digital tickets

Medium

Feature

Description

Priority

Cancellation

Cancel reservations with refund

Medium

Admin Management Add/update buses, fares, schedules High

4. External Interface Requirements

🕐 User Interfaces:

o Web portal with responsive design

o Mobile app interface

🕐 Hardware Interfaces:

o Server hosting, payment gateway integration

🕐 Software Interfaces:

o Database, payment APIs, notification services

🕐 Communication Interfaces:

o Internet connectivity, SMS/email services

5. Non-Functional Requirements

🕐 Performance: Real-time seat updates, <2 sec response time.

🕐 Security: SSL encryption, secure login, payment protection.

🕐 Reliability: 99.9% uptime, backup and recovery.

🕐 Usability: Simple UI, multilingual support.

🕐 Scalability: Support for increasing users and buses.

6. Other Requirements

🕐 Audit logs for transactions

🕐 Compliance with transport authority rules

🕐 Backup and disaster recovery plan